

Quiz 6

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given the following initial value problem

$$\frac{dy}{dx} = y + x + 1, \quad y(0) = -1.$$

Apply **Euler's method** to approximate the solution of the problem on the interval $[0, 0.3]$ with step size $h = 0.1$.

$$\underline{x_0 = 0, h = 0.1, y_0 = -1, f(x, y) = y + x + 1}$$

$$n=1: x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$\begin{aligned} y_1 &= y_0 + h f(x_0, y_0) = y_0 + h(y_0 + x_0 + 1) \\ &= -1 + (0.1)(-1 + 0 + 1) = -1 \end{aligned}$$

$$n=2: x_2 = x_1 + h = 0.1 + 0.1 = 0.2$$

$$y_2 = y_1 + h f(x_1, y_1) = -1 + (0.1)(-1 + 0.1 + 1) = -1 + 0.01 = -0.99$$

$$n=3: x_3 = x_2 + h = 0.2 + 0.1 = 0.3$$

$$\begin{aligned} y_3 &= y_2 + h f(x_2, y_2) = -0.99 + (0.1)(-0.99 + 0.2 + 1) \\ &= -0.99 + (0.1)(0.21) = -0.99 + 0.021 = -0.969 \end{aligned}$$

Problem 2. (5 points) Given the following homogeneous second-order differential equation:

$$4y'' + 8y' + 3y = 0.$$

Verify that both $y_1(x) = e^{-x/2}$ and $y_2(x) = e^{-3x/2}$ satisfy the above equation and the Wronskian $W(y_1, y_2) \neq 0$ for any x . Thus, the general solution of the equation is given by

$$y(x) = c_1 e^{-x/2} + c_2 e^{-3x/2},$$

where c_1 and c_2 are two arbitrary constants.

$$\begin{aligned} y_1(x) &= e^{-x/2}, \quad y_1'(x) = -\frac{1}{2}e^{-x/2}, \quad y_1''(x) = \frac{1}{4}e^{-x/2} \\ 4y_1'' + 8y_1' + 3y_1 &= 4\left(\frac{1}{4}e^{-x/2}\right) + 8\left(-\frac{1}{2}e^{-x/2}\right) + 3e^{-x/2} \\ &= e^{-x/2}(1 - 4 + 3) = 0 \\ y_2(x) &= e^{-3x/2}, \quad y_2'(x) = -\frac{3}{2}e^{-3x/2}, \quad y_2''(x) = \frac{9}{4}e^{-3x/2} \\ 4y_2'' + 8y_2' + 3y_2 &= 4\left(\frac{9}{4}e^{-3x/2}\right) + 8\left(-\frac{3}{2}e^{-3x/2}\right) + 3e^{-3x/2} \\ &= e^{-3x/2}(9 - 12 + 3) = 0 \\ W(y_1, y_2) &= \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x/2} & e^{-3x/2} \\ -\frac{1}{2}e^{-x/2} & -\frac{3}{2}e^{-3x/2} \end{vmatrix} \\ &= (e^{-x/2})\left(-\frac{3}{2}e^{-3x/2}\right) - \left(-\frac{1}{2}e^{-x/2}\right)(e^{-3x/2}) \\ &= \left(-\frac{3}{2} + \frac{1}{2}\right)e^{-x/2}e^{-3x/2} \\ &= -e^{-2x} \neq 0 \text{ for any } x. \end{aligned}$$